

Precision Matrix Decomposition: The Relational Theory

Λ_{zz}	Λ_{zx}
Λ_{xz}	Λ_{xx}

Λ (total)

=

Λ_{zz}	0
0	Λ_{xx}

D (nodal)

+

0	Λ_{zx}
Λ_{xz}	0

R (relational)

MFVI keeps D

(nodal structure)

MFVI discards R

(relational structure)

Core Insight: Inference Coupling (IC) measures what MFVI discards

$$IC = |\Lambda_{zx}| / \sqrt{(\Lambda_{zz} \cdot \Lambda_{xx})} = |\rho| \quad (\text{for bivariate Gaussian})$$